

Week 4 - assignment

Title: Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

Task 1:

- Explore Azure Portal
- Create a Resource Group

The screenshot displays the Microsoft Azure portal interface. The top navigation bar includes the 'Microsoft Azure' logo, a search bar, and user information for '11232554_anuj@mmu...'. The main content area is titled 'Resource Manager | Resource groups'. On the left, a sidebar lists navigation options: Resource Manager, All resources, Favorite resources, Recent resources, Resource groups (selected), Tags, Organization, Tools, Deployments, and Help. The main panel shows a table of resource groups with one entry: 'week4CEIAnuj' under the 'Azure for Students' subscription in the 'East Asia' location. Above the table, there are filters for 'Subscription equals all' and 'Location equals all', along with buttons for 'Create', 'Manage view', 'Refresh', 'Export to CSV', 'Open query', 'Assign tags', 'Add to service group', and 'Add filter'. At the bottom, it indicates 'Showing 1 - 1 of 1. Display count: auto' and a 'Give feedback' link.

Name ↑	Subscription	Location
week4CEIAnuj	Azure for Students	East Asia

Task 2: Storage Setup

- Create a Storage Account
- Create a Blob Container
- Upload a CSV file

The screenshot shows the Microsoft Azure portal interface. At the top, there's a blue header with the Microsoft Azure logo, a search bar, and user information. Below the header, the main content area displays the 'Overview' page for a deployment named 'week4anuj485_1781702866158'. The deployment status is 'Complete', indicated by a green checkmark. The deployment details show it was created on 6/17/2026 at 6:57:09 PM, using the 'Azure for Students' subscription and the 'week4CEIAnuj' resource group. The left sidebar contains a navigation menu with 'Overview', 'Inputs', 'Outputs', and 'Template'. The right sidebar features several recommendations: 'Cost management' (Get notified to stay within your budget), 'Microsoft Defender for Cloud' (Secure your apps and infrastructure), 'Free Microsoft tutorials' (Start learning today), and 'Work with an expert' (Find an Azure expert).

Home

Microsoft Azure

Search resources, services, and docs (G+)

11232554_anuj@mmu...
DEFAULT DIRECTORY (11232554A...

week4anuj485_1781702866158 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : week4anuj485_1781702866158 Start time : 6/17/2026, 6:57:09 PM
Subscription : Azure for Students Correlation ID : 854c30c6-8682-48f7-be84-452...
Resource group : week4CEIAnuj

> Deployment details

< Next steps

Go to resource

Cost management

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Home > week4anuj485

week4anuj485

Containers

☆

...

Storage account

×

Search

◊

«

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Containers

Classic file shares

Queues

Tables

Security & networking

Add or remove favorites by pressing Ctrl+Shift+F

+ Add container

↑ Upload

↻ Refresh

🗑 Delete

🔒 Change access level

🔄 Restore containers

⚙ Edit columns

Search containers by prefix

Only show active containers

Showing all 3 items

☐	Name	Last modified	Anonymous access level	Lease state	Storage connector	
☐	📁 \$logs	6/17/2026, 6:57:54 PM	Private	Available	-	...
☐	📁 destination-container	6/17/2026, 9:55:29 PM	Private	Available	-	...
☐	📁 source-container	6/17/2026, 9:55:02 PM	Private	Available	-	...

Home > week4anuj485 | Containers

source-container

...

Container

×

Search

◊

«

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

+ Add Directory

↑ Upload

🔒 Change access level

↻ Refresh

🗑 Delete

📄 Copy

📄 Paste

🔄 Rename

🔗 Acquire lease

...

source-container

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state	
☐	📄 Sample - Superst...	6/17/2026, 9:56:12 PM	Hot (Inferred)	Block blob	2.18 MiB	Available	...

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Home

CreateDataFactory-20260617215959 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

- Inputs
- Outputs
- Template

Your deployment is complete

Deployment name : CreateDataFactory-202606172... Start time : 6/17/2026, 10:04:08 PM
Subscription : Azure for Students Correlation ID : 6603169d-a830-42ed-b660-f2d...
Resource group : week4CEIAnuj

> Deployment details

< Next steps

Go to resource

Deployment succeeded
Deployment 'CreateDataFactory-20260617215959' to resource group 'week4CEIAnuj' was successful.

Pin to dashbo...

Go to resource gr...



Cost management

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Secure your apps and infrastructure
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Work with an expert

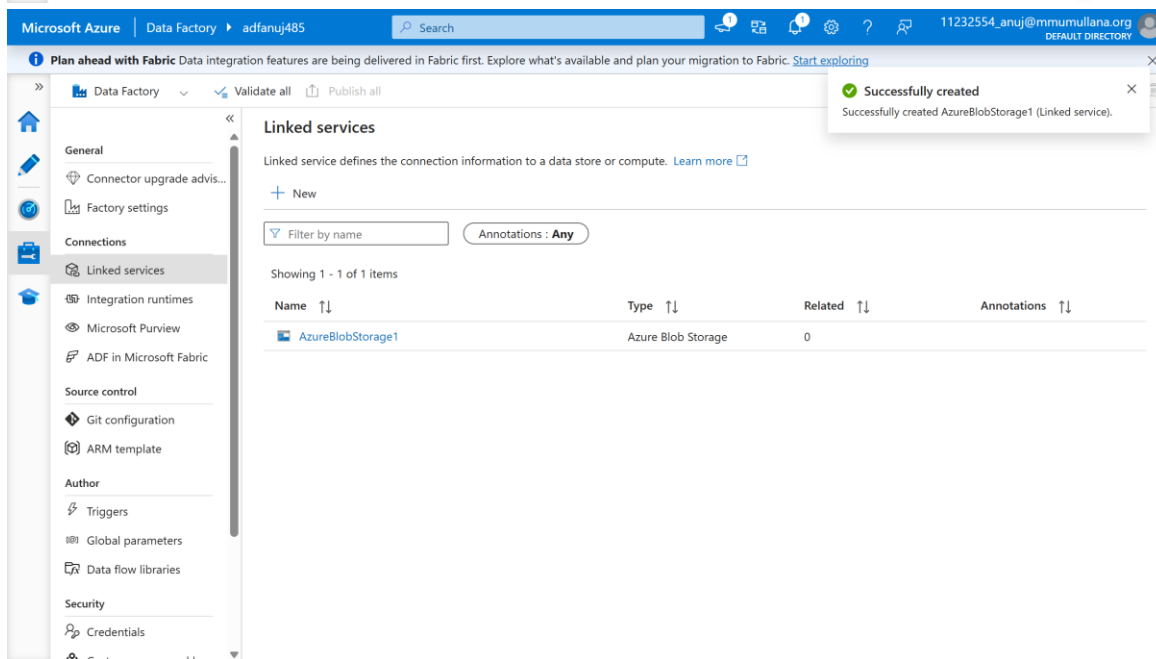
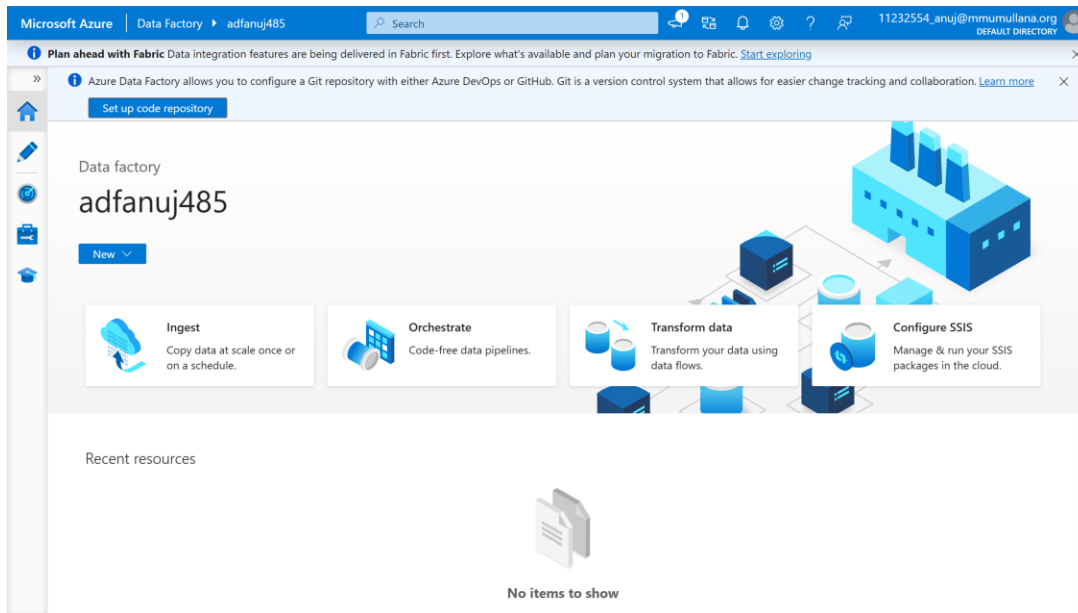
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Task 3: ADF Basics

- Create Azure Data Factory
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)
- Create datasets (source + destination)
- Use Get Metadata activity



Microsoft Azure | Data Factory | adfanuj485 | Search | 11232554_anuj@mmumullana.org | 11232554_anuj@mmumullana.org | DEFAULT DIRECTORY

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

» Data Factory | Validate all | Publish all | Migrate to Fabric (Preview)

Factory Resources

Filter resources by name

- Pipelines 1
 - BlobCopyPipeline
- Change Data Capture (preview) 0
- Datasets 2
 - DelimitedText1
 - DelimitedText2
- Data flows 0
- Power Query 0

Activities

Search activities

- Databricks
- General
 - Append variable
 - Delete
 - Execute Pipeline
 - Execute SSIS package
 - Fail
 - Get Metadata
 - Lookup
 - Stored procedure
 - Script
 - Set variable
 - Validation

Get Metadata

Get Source File Metadata

Properties

General Related

Name *
BlobCopyPipeline

Description

Annotations
+ New

Parameters Variables Settings Output

Pipeline 1a23dc7f-e4ff-44fa-be
run ID 5a-a9980823277c

Pipeline status **Succeeded**
[View debug run consumption](#)

All status Monitor in Azure Metrics Export to CSV

Showing 1 - 1 of 1 item

Activity name Activity status

Get Source File Metadata Succeeded

Cancel

Task 4: Pipeline Development

- Create pipeline using Copy Data activity
- Configure source and destination
- Add ForEach activity (optional if multiple files)

The screenshot shows the Microsoft Azure Data Factory console. On the left, the 'Factory Resources' pane lists 'Pipelines' and 'Datasets'. The 'Pipelines' section shows 'BlobCopyPipeline' selected. The 'Activities' pane on the right shows a 'Copy data' activity configured with 'DelimitedText2' as the sink dataset. The 'Properties' pane on the far right shows the 'General' tab with the pipeline name 'BlobCopyPipeline' and a description field. The main canvas shows a visual representation of the pipeline with a 'Get Metadata' activity followed by a 'Copy data' activity.

The screenshot shows the Microsoft Azure Storage Explorer interface. The 'destination-container' is selected, and the 'Overview' tab is active. The 'Authentication method' is set to 'Access key'. The 'Add filter' button is visible. The 'Search blobs by prefix (case-sensitive)' field is empty. The 'Only show active blobs' dropdown is set to 'Only show active blobs'. The table below shows the contents of the container:

Name	Last modified	Access tier	Blob type	Size	Lease state
Sample - Superst...	6/17/2026, 10:53:01 PM	Hot (Inferred)	Block blob	2.18 MiB	Available

Task 5: Pipeline Execution

- Run pipeline using Debug/Trigger

Microsoft Azure | Data Factory | adfanuj485

Search

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Validate all Publish all

Migrate to Fabric (Preview)

Factory Resources

Filter resources by name

Pipelines 1

BlobCopyPipeline

Change Data Capture (preview) 0

Datasets 2

DelimitedText1

DelimitedText2

Data flows 0

Power Query 0

Activities

Search activities

Move and transform

Copy data

Data flow

Synapse

Azure Data Explorer

Azure Function

Batch Service

Databricks

General

HDInsight

Iteration & conditionals

Machine Learning

Power Query

Get Metadata

Get Source File Metadata

Copy data

Copy data1

Properties

General Related

Name *

BlobCopyPipeline

Description

Annotations

+ New

Cancel

Parameters Variables Settings Output

Pipeline 5578911b-44d6-4cf0-b6fe-61fc0feab8b6

run ID 6fe-61fc0feab8b6

Pipeline status Succeeded

View debug run consumption

All status Monitor in Azure Metrics Export to CSV

Showing 1 - 2 of 2 items

Activity name	Activity st...	Acti
Copy data1	Succeeded	Cop
Get Source File Metadata	Succeeded	Get

Task 6: IAM Roles

- Assign roles:
 - Reader
 - Contributor
- Provide access to ADF for Storage

Home > Storage center | Blob Storage > week4anuj485

week4anuj485 | Access Control (IAM)

☆

...

Storage account

IAM

×

×

«

+

Add

↓

Download role assignments

≡

Edit columns

↻

Refresh

🗑

Delete

🗨

Feedback

Access Control (IAM)

Check access

Role assignments

Roles

Deny assignments

Number of role assignments for this subscription ⓘ

4 4000

Search by name or email

Type: All

Role: All

Scope: All scopes

Group by: Role

All (4) Job function roles (2) Privileged administrator roles (2)

☐	Name ↑↓	Type ↑↓	Role ↑↓	Scope ↑↓	Condition ↑↓
▼ Owner (2)					
☐	AK Anuj Kumar 5ff03381-a4f8-4971...	User	Owner	Subscription (Inherited)	None
☐	AK Anuj Kumar 5ff03381-a4f8-4971...	User	Owner	Subscription (Inherited)	None
▼ Reader (1)					
☐	adfanuj485 9eb931b1-a7df-45...	Managed identity	Reader	This resource	None
▼ Storage Blob Data Contributor (1)					
☐	adfanuj485 9eb931b1-a7df-45...	Managed identity	Storage Blob Data Contrib...	This resource	Add

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